

Newton decomposition of tame character sheaves on loop groups

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Lecture 1. G (split) reductive group / $k = \bar{k}$

$$(B \supset U, B \twoheadrightarrow T)$$

$$B \backslash G / B \leftarrow \frac{U \backslash G / U}{\text{Ad } T} \xleftarrow{\delta} \frac{G}{\text{Ad } B} \xrightarrow{\pi} \frac{G}{\text{Ad } G}$$

$$CS^{\text{unip}}(G) = \langle \pi! \delta^* F : F \in D(B \backslash G / B) \rangle$$

P parabolic $\twoheadrightarrow L_P$

$$\frac{L_P}{L_P} \xleftarrow{q} \frac{P}{P} \xrightarrow{r} \frac{G}{G}$$

$$CS'^{\text{unip}}(G) = \langle \pi! q^* A : A \in CS^{\phi}(L_P) \rangle$$

$$CS = CS'.$$

}

L_G loop group of G , $L_G(k) = G(k((t)))$

$$I \supset I^+, I \xrightarrow{\theta} T$$

$\theta \in L_G$ automorphism preserving I, T

$$\frac{I^+ \backslash L_G / I^+}{\text{Ad}_\theta T} \xleftarrow{\delta} \frac{L_G}{\text{Ad}_\theta I} \xrightarrow{\pi} \frac{L_G}{\text{Ad}_\theta L_G} \quad \text{Ad}_\theta(g) \cdot g' = g g' \theta(g)^{-1}$$

$$\text{Def 1. } CS_{\text{t.g.}}^{\text{tame}}(L_G, \theta) = \left\langle \pi_* \delta^! F : \begin{array}{l} F \in D_c^b\left(\frac{I^+ \backslash L_G / I^+}{T}\right) \\ \text{mono-dromic along } \frac{I^+ \backslash L_G / I^+}{T} \rightarrow I \backslash L_G / I \end{array} \right\rangle$$

$\mathcal{P}$ parahoric $\subset LG$

$\downarrow$

$L_{\mathcal{P}}$ - f.d. reductive gp / k

$$\frac{L_{\mathcal{P}}}{\text{Ad}_{\mathcal{P}} L_{\mathcal{P}}} \xleftarrow{\varphi} \frac{\mathcal{P}}{\text{Ad}_{\mathcal{P}} \mathcal{P}} \xrightarrow{\gamma} \frac{LG}{\text{Ad}_{\mathcal{P}} LG}$$

$$CS_{\text{f.g.}}^{\text{tame}}(LG_{\{1\}}, \theta) = \langle r_* q^! A : A \in CS^{\phi}(L_{\mathcal{P}}, \theta) \rangle$$

To explain the relation between $CS^{\text{tame}}(LG, \theta)$, $CS^{\text{tame}}(LG_{\{1\}}, \theta)$,

we discuss some geom. of $\frac{LG}{\text{Ad}_{\mathcal{P}} LG}$.

Fact. $\exists$ a partially ordered subset

$$N(G, \theta) \hookrightarrow X_*(T)_{\mathcal{O}_1}^+ \times \pi_1(G) \leftarrow \text{partially ordered}$$

$$\downarrow \quad (v, \varphi) \quad (v', \varphi') \leq (v, \varphi)$$

$$\vartheta \longmapsto (v_{\vartheta}, k_{\vartheta})$$

if $\varphi' = \varphi$, $v - v'$ is a positive rat'l combination of simple roots.

such that - for every ϑ ,

$\{\vartheta' : \vartheta' \leq \vartheta\}$ is finite,

- minimal elements (called basic)

$$N(G, \theta)_{\mathcal{O}} \xrightarrow{\sim} \pi_1(G)$$

Th 1. (1) $\exists$ cts map

$$\left(\frac{L_{\mathcal{G}}}{\text{Ad}_{\mathcal{G}} L_{\mathcal{G}_{\text{red}}}} \right) \longrightarrow \mathcal{N}(\mathcal{G}, \mathcal{O})$$

$$\frac{L_{\mathcal{G}\mathcal{O}}}{\text{Ad}_{\mathcal{G}} L_{\mathcal{G}_{\text{red}}}} \longmapsto \mathcal{O}$$

(2) $\frac{L_{\mathcal{G} \leq \mathcal{O}}}{\text{Ad}_{\mathcal{G}} L_{\mathcal{G}_{\text{red}}}} \xrightarrow{L \leq \mathcal{O}} \frac{L_{\mathcal{G}_{\text{red}}}}{\text{Ad}_{\mathcal{G}} L_{\mathcal{G}_{\text{red}}}}$ is finitely presented closed embedding.

$\frac{L_{\mathcal{G}\mathcal{O}}}{\text{Ad}_{\mathcal{G}} L_{\mathcal{G}_{\text{red}}}} \xrightarrow{j_{\mathcal{O}}} \frac{L_{\mathcal{G} \leq \mathcal{O}}}{\text{Ad}_{\mathcal{G}} L_{\mathcal{G}_{\text{red}}}}$ is quasi-compact open embedding

(3) $\mathcal{O} \rightsquigarrow (M = \mathbb{Z}_\ell(v_{\mathcal{O}}), \mathcal{O}_M \in \mathcal{N}(M, \mathcal{O}) \text{ basic})$

$$\left(\frac{L_{\mathcal{G}\mathcal{O}}}{\text{Ad}_{\mathcal{G}} L_{\mathcal{G}_{\text{red}}}} \right)_h \xleftarrow{\sim} \left(\frac{L M \mathcal{O}_M}{\text{Ad}_{\mathcal{G}} L M_{\text{red}}} \right)_h$$

(4) If $\mathcal{O}$ is basic, $\exists$ "nice" $x \in L_{\mathcal{G}\mathcal{O}}$ st

$$L_{\mathcal{G}\{x\}} \xleftarrow{\sim} L_{\mathcal{G}\mathcal{O}}$$

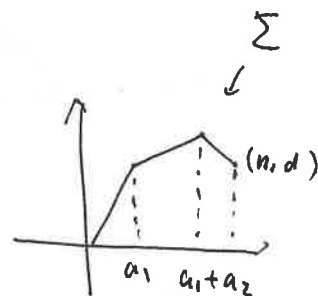
$$g.x \longleftarrow g$$

$$\hookrightarrow \hookrightarrow$$

$$\text{Ad}_x \circ \theta =: \theta_x$$

$$\theta$$

Ex. $\mathcal{G} = \mathcal{G}_n$, $\theta = \text{id}$, $\mathcal{N}(\mathcal{G}, \theta) := \left\{ \begin{array}{l} \text{concave Newton polygon} \\ (0,0), (n,d) \end{array} \right\}$



slope ↓
 $\left(\left(\frac{x_1, \dots, x_1}{a_1}, \frac{x_2, \dots, x_2}{a_2}, \dots \right), d \right) \in X_*(T)^+_{a_1} \times \pi_2(\mathcal{G})$

$\Sigma \leq \Sigma'$ iff Σ, Σ' have the same end pt & Σ' is above Σ

Σ basic $\Leftrightarrow \Sigma$ is a straight line.

$$LG \ni A \rightsquigarrow \text{char}_T(A) = T^n + a_1 T^{n-1} + \dots + a_n$$

Σ is then
concave hull of $\{(i, \text{val}(a_i))\}$

$$\emptyset = \{1\}, \quad LG_{\{1\}} = \{A : \text{char}_T(A) \in k[[\infty]] [T]\}$$

$\xrightarrow{\quad}$

Def. (1) $\emptyset$ basic, $CS_{f.g.}^{\text{tame}}(LG_{\emptyset}, \emptyset) := CS_{f.g.}^{\text{tame}}(LG_{\{1\}}, \emptyset_x) \subset D\left(\frac{LG_{\emptyset}}{\text{Ad}_{\emptyset} LG_{\text{red}}}\right)$

(2) $\emptyset$ general, $CS_{f.g.}^{\text{tame}}(LG_{\emptyset}, \emptyset) := CS_{f.g.}^{\text{tame}}(LM_{\emptyset_M}, \emptyset)$

Th. The following subcat. of $D\left(\frac{LG}{\text{Ad}_{\emptyset} LG}\right)$ are the same:

(1) $CS_{f.g.}^{\text{tame}}(LG, \emptyset)$

(2) $\left\{ F \in D\left(\frac{LG}{\text{Ad}_{\emptyset} LG}\right) : \begin{array}{l} (2\emptyset)^! F \in CS_{f.g.}^{\text{tame}}(LG_{\emptyset}, \emptyset) \text{ for every } \emptyset, \\ \text{and } (2\emptyset)^! F = 0 \text{ for almost all } \emptyset \end{array} \right\}$

(3) $\left\{ F \in D\left(\frac{LG}{\text{Ad}_{\emptyset} LG}\right) : \begin{array}{l} (2\emptyset)^* F \in CS_{f.g.}^{\text{tame}}(LG_{\emptyset}, \emptyset) \text{ for every } \emptyset \\ \text{and } (2\emptyset)^* F = 0 \text{ for almost all } \emptyset \end{array} \right\}$

Cor. $CS_{f.g.}^{\text{tame}}(LG_{<\emptyset}, \emptyset) \xrightarrow[\leftarrow (2\emptyset)^!]{\leftarrow (2\emptyset)^*} CS_{f.g.}^{\text{tame}}(LG_{\leq \emptyset}, \emptyset) \xrightarrow[\leftarrow (2\emptyset)^*]{\leftarrow (2\emptyset)^!} CS_{f.g.}^{\text{tame}}(LG_{\emptyset}, \emptyset)$

$$\underline{\text{Th.}} \quad (S_{\text{t.g.}}^{\text{tame}}(L_{G\{1\}}, \theta) = \bigoplus_{(P, A)/\sim} \langle (r_P)_* q_P^! A \rangle$$

$$P \text{ parahoric, } A \in CS_c^\phi(L_P, \theta)$$

One can define a t-str. on $(S_{\text{t.g.}}^{\text{tame}}(L_{G\{1\}}, \theta)$

$$\text{Ind}_{CS^{\text{tame}}} (S_{\text{t.g.}}^{\text{tame}}(L_{G\{1\}}, \theta)^{\leq 0} := \left\{ (r_P)_* (q_P)^! F : F \in D_c(L_P)^{\leq 0}, F \in (S_c(L_P)) \right\}$$

Then $(r_P)_* q_P^!$ is t-exact

BKV $\downarrow$ should be t-exact

$$D\left(\frac{L_{G\{1\}}^{\text{rss}}}{A \in L_G}\right)$$

A cusp,

irr. rep.

$$(r_P)_* (q_P)^! A \otimes \bigvee \text{ is in the heart.}$$

$H^0 \text{End}(\dots)$

Lecture 2

$$\frac{I^+ \backslash L_G / I^+}{T} \xleftarrow{\delta} \frac{L_G}{I} \xrightarrow{\pi} \frac{L_G}{L_G}, \quad \theta = \text{id}$$

$$(S_{\text{t.g.}}^{\text{tame}}(L_G) = \langle \pi_* \delta^! F : F \in D_{\text{f.g. mon}}^b(I^+ \backslash L_G / I^+) \rangle$$

$$\left| \frac{L_G}{L_G} \right| \rightarrow r(\lambda) \ni 0 \rightsquigarrow (M, \theta_M)$$

$$\frac{L_{G\theta}}{L_G} \simeq \frac{L_{M\theta_M}}{L_M}$$

$$(S_{\text{t.g.}}(L_{G\theta}) := (S_{\text{t.g.}}(L_{M\theta_M}))$$

$$\frac{L_P}{L_P} \xleftarrow{q} \frac{P}{P} \xrightarrow{r} \frac{L_G}{L_G}$$

$$(S_{\text{t.g.}}(L_{G\{1\}}) = \langle r_* q^! A : A \in CS^\phi(L_P) \rangle$$

pages

Th. The following subcategories of $D(\frac{L_h}{L_h})$ are the same:

(i) $CS_{f.g.}^{tame}(L_h)$

(ii) $\{F \in D(\frac{L_h}{L_h}) : (2_0)^! F \in CS_{f.g.}(L_{h0}) \text{ \& } (2_0)^! F = 0 \text{ for almost all } 0\}$

(iii) $\{F \in D(\frac{L_h}{L_h}) : (2_0)^* F \in CS_{f.g.}(L_{h0}) \text{ \& } (2_0)^* F = 0 \text{ for almost all } 0\}$

Th $\mathcal{H} = \bigoplus_{x, x'} D((I, x) \setminus L_h / (I, x')) \supset \mathcal{H}_{f.g.} = \bigoplus_w D_c^b((I, x) \setminus L_{h,w} / (I, x'))$

$x, x' \in Loc^{mult}(T)$

$$\begin{array}{ccc} Tr(\mathcal{H}_{f.g.}) & \xrightarrow{\quad} & D_{f.g.}(\frac{L_h}{L_h}) \\ & \searrow \sim & \downarrow \\ & CS_{f.g.}^{tame}(L_h) & \\ \downarrow & \nearrow \hookrightarrow & \\ Tr(\mathcal{H}) & \xrightarrow{\quad} & D(\frac{L_h}{L_h}) \quad \square \end{array}$$

Rem. $\frac{L_h}{L_h}$ sifted ind-placid stack

$$\begin{array}{c} \rightarrow \\ \Rightarrow \end{array} \frac{L_h}{I} \times \frac{L_h}{L_h} \Rightarrow \frac{L_h}{I} \Rightarrow \frac{L_h}{L_h}$$

$\mathcal{X}$ sifted placid, $D_{f.g.}(\mathcal{X}) = \text{colim } D_c^b(Y)$
 $Y \rightarrow \mathcal{X}$ Y placid stack
ind t.p.

$\exists D_{f.g.}(\mathcal{X}) \rightarrow D(\mathcal{X})$, not fully faithful in general

eg. $\mathcal{X} = IB L_h \Rightarrow IB L^+ L_h \xrightarrow{\neq} IB L_h$

$$\text{End}_{D_{\text{fg}}}(\text{IBL}_G) \left(\overset{A_{\text{f.g.}}}{\pi_* \omega_{\text{IBL}^+ G}} \right)$$

$$\text{End}_{D(\text{IBL}_G)} \left(\overset{A}{\pi_* \omega_{\text{IBL}^+ G}} \right)$$

$$L^+ G \backslash L_G / L^+ G \xrightarrow{p_2} \text{IBL}^+ G$$

$$\begin{array}{ccc} p_1 \downarrow & & \downarrow \pi \\ \text{IBL}^+ G & \xrightarrow{\pi} & \text{IBL}_G \end{array}$$

$$\begin{aligned} & \text{End}_{\left(\overset{A}{\text{f.g.}} \right)} \\ &= \text{Hom}_{\text{Ind } D_G^b(\text{IBL}^+ G)} \left(\omega, p_{L*} \omega_{L^+ G \backslash L_G / L^+ G} \right) \\ & \quad D(\text{IBL}^+ G) \end{aligned}$$

$$= \left[\begin{array}{l} \text{equiv. Bond-Moore homology} \\ \text{of } L_G \end{array} \right]$$



Let $\tilde{W}$ be the (extended) affine Weyl group

$$\begin{array}{c} \downarrow \\ \text{Waff} \\ \downarrow \\ 1 \rightarrow X_*(T) \rightarrow \tilde{W} \rightarrow W \rightarrow 1 \\ \downarrow \nearrow \\ \pi_1(G) \cong \Omega = \{ w \in \tilde{W} : w I w^{-1} = I \} \\ \downarrow \\ 1 \end{array}$$

$$\tilde{W} \longrightarrow X_*(T)_G \times \pi_1(G)$$

$$w \longmapsto (\tilde{\nu}_w, \kappa_w)$$

$$w^n = n \tilde{\nu}_w \quad n \text{ sufficiently divisible (by } |w|)$$

$$\tilde{\nu}_{xwx^{-1}} = x \tilde{\nu}_w x^{-1}, \quad \kappa_{xwx^{-1}} = \kappa_w$$

$$\frac{\tilde{w}}{\tilde{w}} \longrightarrow X_*(T)_{\mathbb{Q}}^+ \times \pi_1(G)$$

$$\searrow \quad \nearrow$$

$$r(G)$$

$$\begin{array}{ccc} \frac{L_G}{I} & & \tilde{w} \\ \downarrow & & \downarrow \\ \frac{L_G}{L_G} & & r(G) \end{array}$$

Def w is straight if

$$l(w^n) = n l(w), \quad \forall n \quad (\Rightarrow l(w) = \langle 2\rho, \nu_w \rangle)$$

A conj. class of $\tilde{w}$ is called straight if it contains a straight element.

$$\text{Th (He) (1)} \quad \left(\frac{\tilde{w}}{\tilde{w}} \right)_{\text{str}} \Rightarrow r(G)$$

$$\quad \quad \quad \emptyset \quad \quad \quad \emptyset$$

$$(2) \quad \emptyset' \leq \emptyset \quad \text{if } \exists \underbrace{w' \in \emptyset', u \in \emptyset}_{\text{str.}}$$

$$w' \leq w \quad \text{in the Bruhat order}$$

Construction of $\frac{L_{G, \emptyset}}{L_G}$

Step 1. Pick w straight in $\emptyset$

Step 2. Independent of the choice of w

$$\begin{array}{ccc} \frac{\overline{IwI}}{I} & \hookrightarrow & \frac{L_G}{I} \\ \downarrow & & \downarrow \\ \text{Define } \frac{L_{G \leq \emptyset}}{L_G} & \longrightarrow & \frac{L_G}{L_G} \end{array}$$

as the scheme theoretic
image of $\frac{\overline{IwI}}{I}$

$$\text{Step 3. } \frac{L_{G, \emptyset}}{L_G} = \frac{L_{G \leq \emptyset}}{L_G} - \bigcup_{\emptyset' < \emptyset} \frac{L_{G \leq \emptyset'}}{L_G}$$

Def An admissible triple (x, K, C) consists of

- x straight
- $K \subset \mathcal{S}_0 = \{\text{affine simple reflections}\}$ s.t. W_K is finite
- & x is minimal length $W_K \setminus \tilde{W} / W_K$, $x W_K x^{-1} = W_K$
- C is an elliptic Ad_x -conj. class in W_K

Th. For every $(x, K, C, u \in C)$

Let $M = \mathbb{Z}_q(V_x)$. Then $\exists$ an admissible triple $(x', K', C', u' \in C')$ of M

$$\begin{array}{ccc}
 \text{s.t.} & \frac{I x u I}{I} & \xleftarrow{\sim} \frac{I_M x' u' I_{M'}}{I_{M'}} \\
 & \downarrow & \downarrow \\
 & \frac{L_{H_0}}{L_G} & \xrightarrow{\sim} \frac{L_{M \cap M}}{L_M}
 \end{array}$$

$\uparrow$
 length 0 for M

